# JACOBIAN EXAMPLE

$$p_x = c\theta_1 \left( c(\theta_2+\theta_3+\theta_4) \cdot a_4 + c(\theta_2+\theta_3) \cdot a_3 + c\theta_2 \cdot a_2 \right)$$

$$dx = \frac{\partial p_x}{\partial \theta_1} \delta\theta_1 + \frac{\partial p_x}{\partial \theta_2} \delta\theta_2 + \frac{\partial p_x}{\partial \theta_3} \delta\theta_3 + \cdots + \frac{\partial p_x}{\partial \theta_6} \delta\theta_6$$

$$\frac{\partial p_x}{\partial \theta_1} = J_{11} = -s\theta_1 \left( c(\theta_2+\theta_3+\theta_4) \cdot a_4 + c(\theta_2+\theta_3) \cdot a_3 + c\theta_2 \cdot a_2 \right)$$

$$\frac{\partial p_x}{\partial \theta_2} = J_{12} = c\theta_1 \left( -s(\theta_2+\theta_3+\theta_4) \cdot a_4 - s(\theta_2+\theta_3) \cdot a_3 - s\theta_2 \cdot a_2 \right).$$

$$\frac{\partial p_x}{\partial \theta_3} = J_{13} = c\theta_1 \left( -s(\theta_2+\theta_3+\theta_4) \cdot a_4 - s(\theta_2+\theta_3) \cdot a_3 \right)$$

$$\frac{\partial p_x}{\partial \theta_4} = J_{14} = c\theta_1 \left( -s(\theta_2+\theta_3+\theta_4) \cdot a_4 \right)$$

$$\frac{\partial p_x}{\partial \theta_5} = J_{15} = 0$$

$$\frac{\partial p_x}{\partial \theta_6} = J_{16} = 0$$

$$dx = J_{11}\delta\theta_1 + J_{12}\delta\theta_2 + J_{13}\delta\theta_3 + J_{14}\delta\theta_4 + J_{15}\delta\theta_5 + J_{16}\delta\theta_6$$